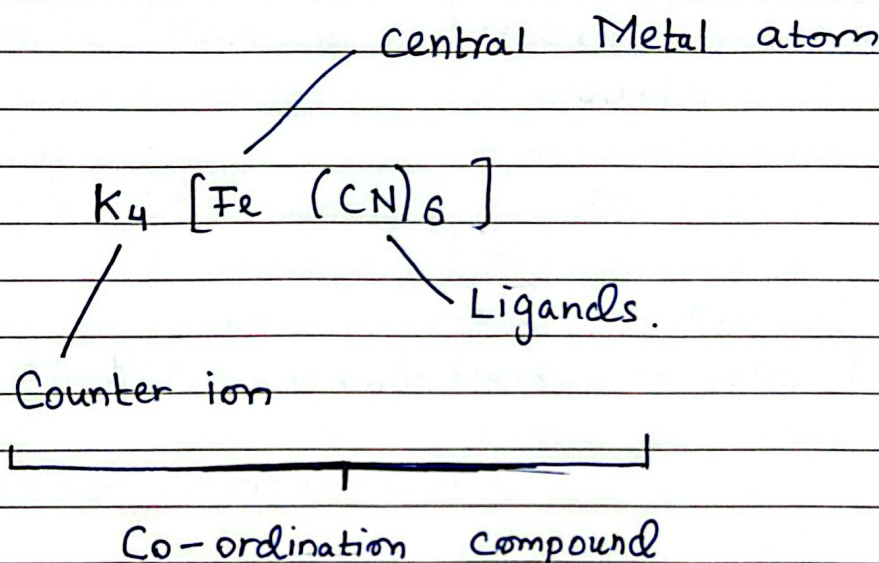


Ltp ~ 5

Coordination Compounds

⇒ IUPAC NAMING



⇒ CLASSIFICATION OF LIGANDS

i) Neutral Ligands : (H_2O) aqua, (NH_3) ammine, (CO) carbonyl, (NO) nitrosyl, $(\text{C}_5\text{H}_5\text{N})$ pyridine.

ii) Anionic ligands : (F^-) Fluorido, (Cl^-) Chlorido, (Br^-) Bromido, (I^-) Iodo, (CN^-) cyanido, (OH^-) Hydroxo, $(\text{CH}_3\text{COO}^-)$ acetato.

iii) Cationic ligands : (NO^+) nitrosylium, $(\text{NH}_2\text{NH}_3^+)$ Hydrazinium.

* Important Unidentate ligands

H_2O - aqua	I^- - iodido
NH_3 - ammine	CN^- - cyanido
CO - Carbonyl	OH^- - Hydroxo
NO - Nitrosyl	NO_2^- - Nitro
C_5H_5N - Pyridine	ONO - nitrito
F^- - Fluorido	CH_3COO^- - acetato
Cl^- - Chlorido	SCN^- - thiocyanato
Br^- - Bromido	NCS^- - isothiocyanato
NH_2^- - Imido	NH_2^- - amido.

* Important Didentate ligands

CH_2NH_2 - ethylenediamine (en)
$C_2O_4^{2-}$ - oxalato.
$EDTA^{4-}$ - ethylenediamine tetraacetate.
$S_2O_3^{2-}$ - thiosulphato

⇒ Rules for IUPAC Naming and Order of Naming

- i] Name of counter ion is written first, no matter if it is before or after parenthesis.
- ii] Ligand name is written in alphabetical order if more than one.
- iii] Followed by the name of central metal atom, then its oxidation number in roman numbers in brackets.
- iv] If the compound has either negative or positive charge "ion" is written after everything at end.
- v] If the entity has a negative charge

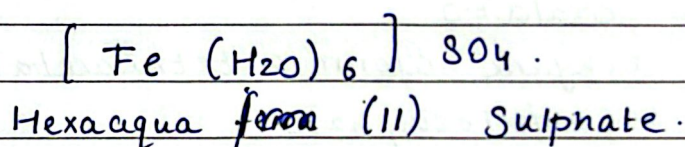
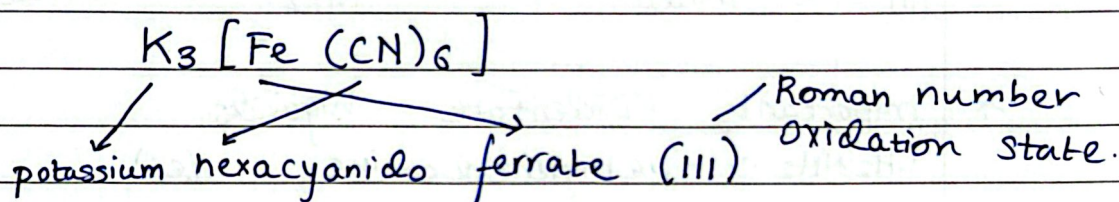
then, the central metal atom should end with 'ate'

Note: If counter ion is

i) before parenthesis then parenthesis is negative and counter ion is positive.

ii) after parenthesis then parenthesis is positive and counter ion is negative.

For Example



⇒ For cationic complex

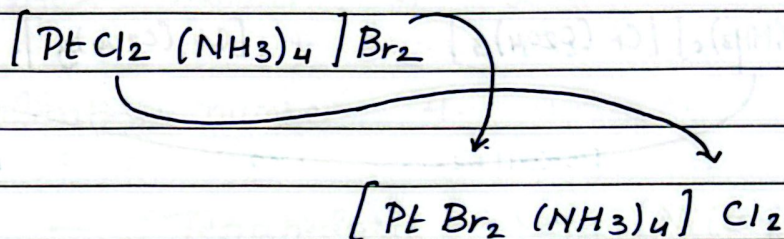
No and name of ligands → Common name of C.M.A → O.S in roman → Name of anion.

⇒ For Anionic complex

Name of Cation → Number and name of ligands → Name of C.M.A with 'ate' → O.S in roman

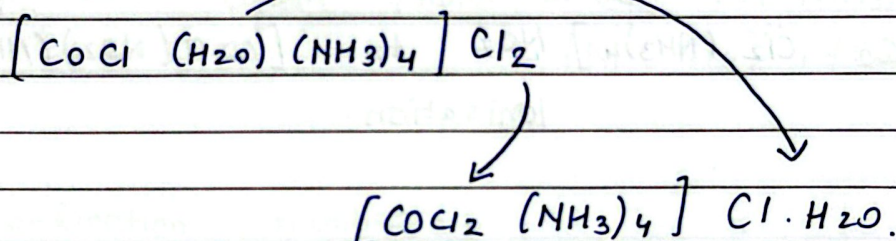
ISOMERISM IN COORDINATION Compounds.

1. Ionisation Isomerism.



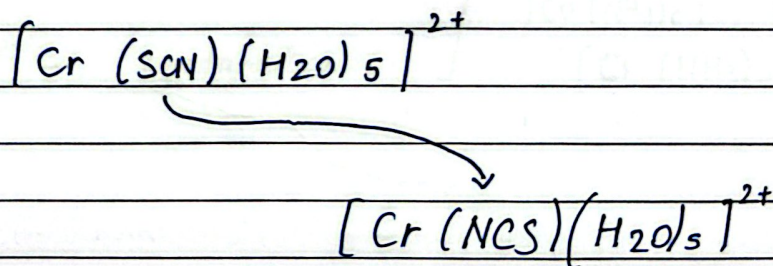
=> Exchange of ligands with counter ion.

2. Hydrate Isomerism.



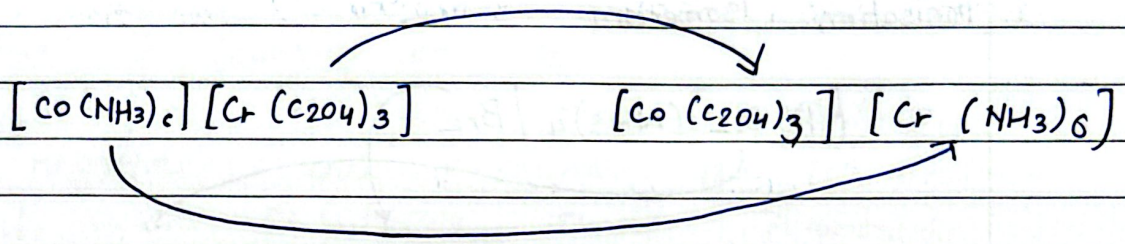
=> The Aqua (water) ligand exchanges outside.

3. Linkage Isomerism

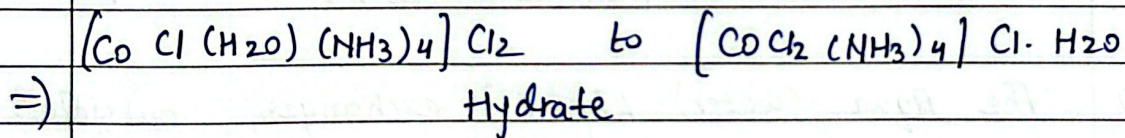
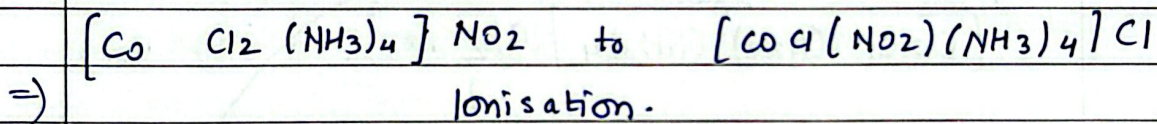
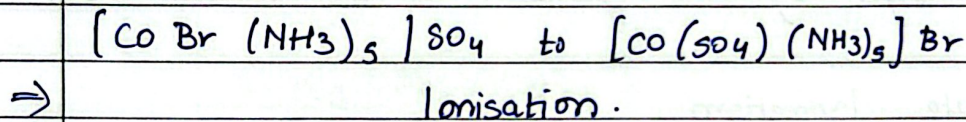


=> The linkage between inside ligands is changed.

4. Coordination isomerism:



Some Examples of isomerism.



$\Rightarrow$

$\Rightarrow$ VALENCE BOND THEORY.

Hybridisation and Geometry based on Coordination number . 4 & 6 .

* Co-ordination number 4.

$\Rightarrow$ sp^3 — Tetrahedral. $\rightarrow$ eg:- $[Ni(CO)_4]$
 $[MnCl_4]^{2-}$, $[ZnCl_4]^{2-}$
 $[NiCl_4]^{2-}$

$\Rightarrow$ dsp^2 — Square Planar — eg:- $[Pt(NH_3)_4]^{2+}$
 $[Ni(CN)_4]^{2-}$, $[Cu(CN)_4]^{2-}$

* Co-ordination number 6 .

$\Rightarrow$ d^2sp^3 — octahedral } eg:- $[Fe(CN)_6]^{4-}$
 $[Fe(CN)_6]^{3-}$
 $[Co(NH_3)_6]^{3+}$, $[PtCl_6]^{2-}$
 sp^3d^2 — octahedral } $[Cr(NH_3)_6]^{3+}$

Steps to Find Hybridisation.

- 1] Identify coordination number of compound
- 2] Then write the electronic configuration of central metal atom.
- 3] Then, after finding the oxidation number of central metal atom, write the electronic configuration of its ion.
- 4] Identify the weak or strong Ligand then correct the electrons by strong attachment or weak attachment.
- 5] Then, fill the electrons of Ligands and find hybridisation.

Note: If the magnetic behaviour is to be found identify if all shells are filled or if they are empty.

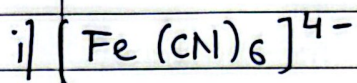
Vacant Shells — Paramagnetic

Empty Shells — Diamagnetic.

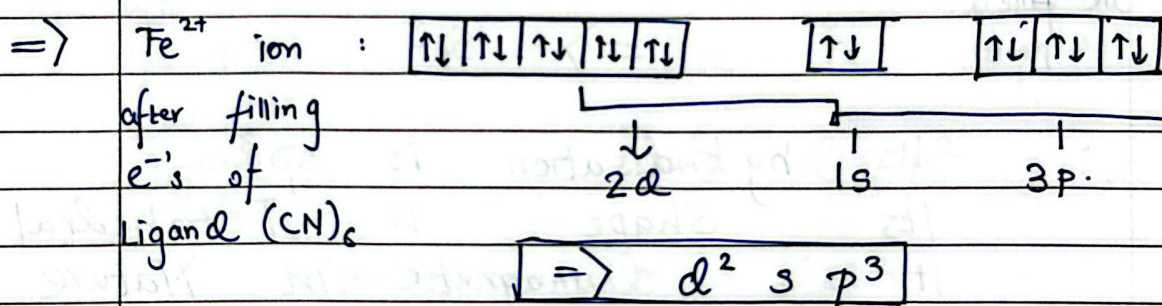
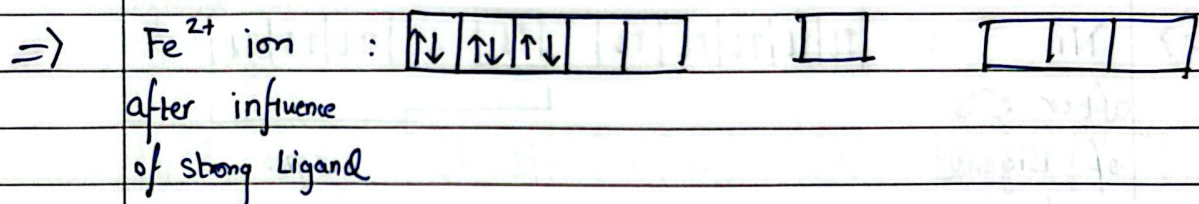
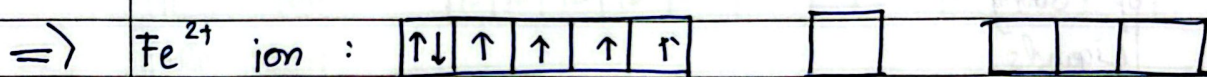
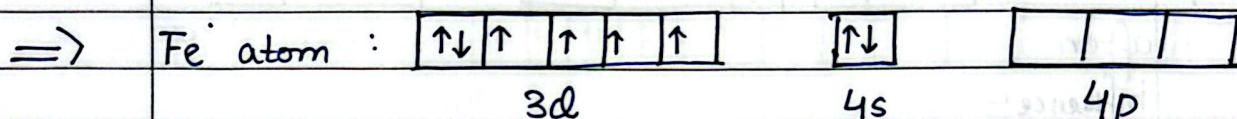
Also, for Spin If.

Weak Ligand — High Spin
Strong Ligand — Low Spin.

Examples for Hybridisation.



$\Rightarrow$ Co-ordination Number = 6.
 $\therefore$ The Shape is Octahedral.



Thus, its hybridisation is $d^2 sp^3$

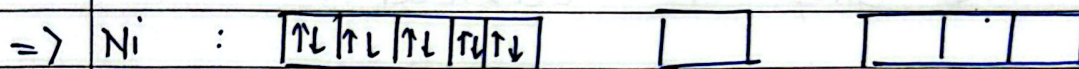
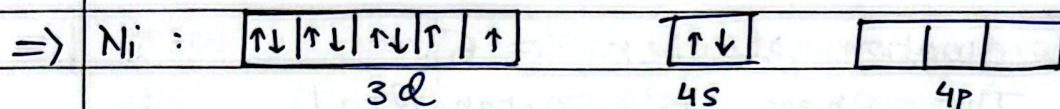
Magnetic Behaviour is Diamagnetic

Spin is Low Spin.

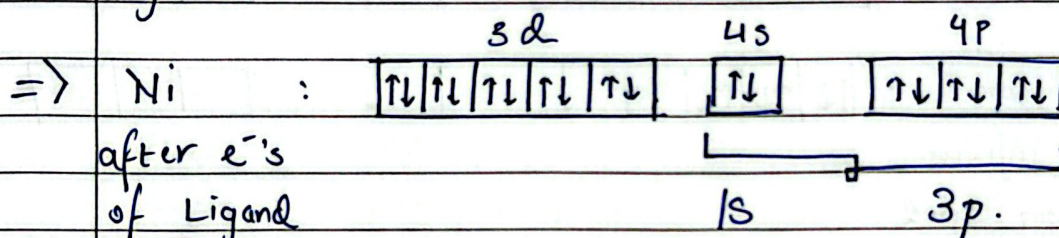
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Example 2 : $[\text{Ni}(\text{CO})_4]$

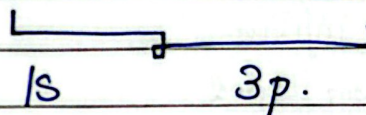
Coordination number : 4.



after
influence
of Strong
Ligands



after e^- 's
of Ligand
are filled
(4 pairs)



$\Rightarrow sp^3$

$\therefore$ Its hybridisation is sp^3

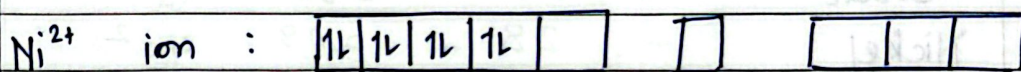
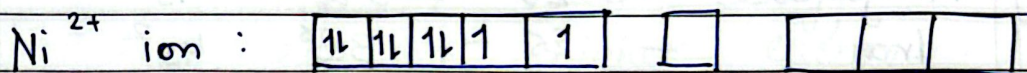
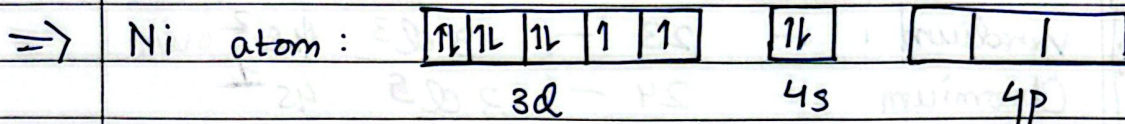
Its Shape is Tetrahedral

It is diamagnetic in Nature

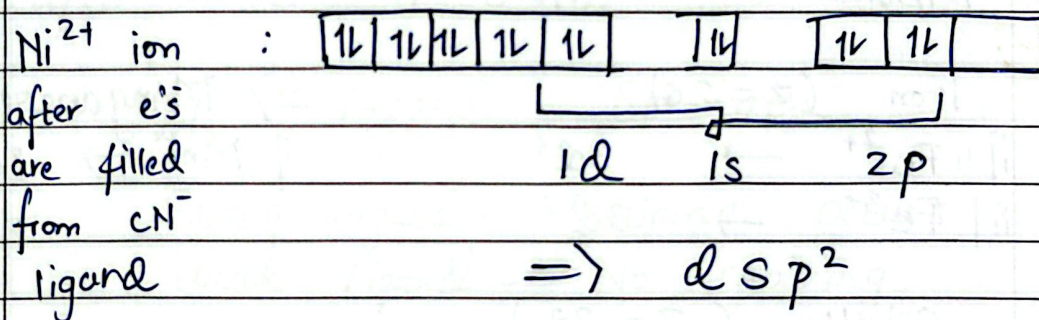
It is Low Spin.

Example 3 : $[\text{Ni}(\text{CN})_4]^{2-}$

Coordination Number : 4.



under strong
influence of
CN⁻ ligands



$\therefore$ Hybridisation = dsp^2
Shape = Square Planar
Paramagnetic and Low Spin

__/11/24

Important Elements Electronic Configuration.

1]	Scandium	—	21	—	$3d^1 4s^2$
2]	Titanium	—	22	—	$3d^2 4s^2$
3]	Vanadium	—	23	—	$3d^3 4s^2$
* 4]	Chromium	—	24	—	$3d^5 4s^1$
5]	Manganese	—	25	—	$3d^5 4s^2$
6]	Iron	—	26	—	$3d^6 4s^2$
7]	Cobalt	—	27	—	$3d^7 4s^2$
8]	Nickel	—	28	—	$3d^8 4s^2$
* 9]	Copper	—	29	—	$3d^{10} 4s^1$
10]	Zinc	—	30	—	$3d^{10} 4s^2$

Important Ions.

$\Rightarrow$ Iron ($Z=26$)

- i] $Fe^{2+} \rightarrow 3d^6$
- ii] $Fe^{3+} \rightarrow 3d^5$

$\Rightarrow$ Manganese ($Z=25$)

- i] $Mn^{2+} \rightarrow 3d^5$

$\Rightarrow$ Cobalt ($Z=27$)

- i] $Co^{2+} \rightarrow 3d^7$
- ii] $Co^{3+} \rightarrow 3d^6$

$\Rightarrow$ Nickel ($Z=28$)

- i] $Ni^{2+} \rightarrow 3d^8$

$\Rightarrow$ Copper ($Z=29$)

- i] $Cu^+ \rightarrow 3d^{10}$
- ii] $Cu^{2+} \rightarrow 3d^9$

$\Rightarrow$ Chromium ($Z=24$)

- i] $Cr^{3+} \rightarrow 3d^3$

CRYSTAL FIELD THEORY.

⇒ the 'd' orbital of coordination compound gets divided into two.

$t_{2g} \rightarrow$ lower energy
 $e_g \rightarrow$ higher energy.

⇒ * How Electrons are filled

i) Write electronic config of C.M.A's ion after removing its oxidation number

ii) Identify Ligand Strength

Strong Ligand - pairing occurs
 Weak Ligand - No pairing.

Example :

⇒ Strong Ligand

t_{2g} $\begin{array}{|c|c|c|} \hline \uparrow\downarrow & \uparrow\downarrow & \uparrow\downarrow \\ \hline \end{array}$

e_g $\begin{array}{|c|c|} \hline & \\ \hline \end{array}$

⇒ weak Ligand

t_{2g} $\begin{array}{|c|c|c|} \hline \uparrow\downarrow & \uparrow & \uparrow \\ \hline \end{array}$

e_g $\begin{array}{|c|c|} \hline \uparrow & \uparrow \\ \hline \end{array}$